

2.3.2 Algorithmic Strategy: Pattern Recognition

پیٹرن ریکگنیشن (Pattern Recognition) کا مطلب ہے ڈیٹا میں دہرائی جانے والی چیزوں یا اصولوں کو پہچاننا۔

CS XI

Chapter 2: Computational Thinking & Algorithm

Lecture 04 of 11

2.3.2 Algorithmic Strategy: Pattern Recognition

LEARN IT

Pattern recognition is the process of identifying regularities, repetitions or rules in data. It helps in reducing the overall complexity of an algorithm and forms the foundation of concepts such as **loops**.

How it works — ask yourself three questions:

- What is repeating?
- What changes in a predictable way?
- Can I generalise this (write one rule that works for every case)?

WORKED EXAMPLE

Example: Triangle of stars

OUTPUT

```
*  
**  
***  
****  
*****
```

Step 1 — What repeats? Every row prints stars.

Step 2 — What changes predictably? Each row adds 1 star to the previous row's count.

Row	Stars
1	1
2	2
3	3
...	...
N	N

Step 3 — Generalise: Row N = N stars. Recognising this pattern lets us write one generic loop instead of 5 separate print statements.

Without the pattern (5 statements)

```
print("*")  
print("**")  
print("***")  
print("****")  
print("*****")
```

With the pattern (one loop)

```
n = 5  
for row in range(1, n + 1):  
    print("*" * row)
```

Both programs print exactly the same triangle. But if you want 100 rows, the loop version needs only one change (`n = 100`), while the first version needs 100 lines.

REMEMBER IT

Whenever you see "the same thing again and again, with a small change each time", think **loop**. Pattern recognition is the habit of spotting that.

COMMON MISTAKE

Do not generalise from too few cases. Check your rule on several rows (or items) before trusting it. "Row N = N stars" is correct here; a different triangle (1, 3, 5, 7 stars) follows the rule "Row N = 2N - 1 stars".

پیٹرن ریکگنیشن (Pattern Recognition) کا مطلب ہے ڈیٹا میں دہرائی جانے والی چیزوں یا اصولوں کو پہچاننا۔ ستاروں کی تکون میں ہر قطار میں پچھلی قطار سے ایک ستارہ زیادہ ہے، یعنی قطار N میں N ستارے۔ یہ پیٹرن پہچان کر ہم پانچ الگ پرنٹ کے بجائے ایک لوپ (Loop) لکھ سکتے ہیں۔

CHECK YOURSELF

1. Which programming concept is based on pattern recognition?
2. How many stars will row 8 of the triangle have?
3. Find the rule: 2, 4, 6, 8, ...

Answers: 1. Loops. 2. 8 stars (Row N = N stars). 3. Term N = 2 × N (each term adds 2).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Pattern recognition is:

- a. Spotting repeats and rules in data
- b. Hiding details
- c. Breaking into parts
- d. A TCP flag

MCQ 2. If every student card has name+photo+id, that is a:

- a. Pattern
- b. Virus
- c. K-map wrap
- d. Sprint

MCQ 3. Patterns let you:

- a. Reuse a solution
- b. Delete the algorithm
- c. Avoid abstraction
- d. Skip testing

MCQ 4. Spam filters look for:

- a. Patterns in text
- b. OSI layer 1 copper
- c. NAND only
- d. Beachhead markets

MCQ 5. No pattern means:

- a. Treat the case as unique
- b. Always bubble sort
- c. Always pie charts
- d. Always Agile

MCQ 6. CT pillar closest to "I've seen this shape of problem" is:

- a. Pattern recognition
- b. Physical layer
- c. Variance
- d. Figma

Extra short questions

1. Define pattern recognition.
2. Give a school example.
3. How do patterns shrink an algorithm?
4. What if you force a pattern that is not there?

Extra long question

1. Show pattern recognition on: (a) attendance lists, (b) weather by month, (c) login screens. For each, name the repeating structure and the algorithm you reuse.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Finding regularities, repetitions or rules so you can reuse a method.
2. Every paper has roll, name, marks — same record shape, same processing loop.
3. One loop or one function handles every similar case.
4. Wrong reuse — the solution fails on the odd case.

Long

1. (a) row per student → linear scan/count. (b) month+value → line graph / average. (c) username+password+submit → same validation function.

Next lecture: 2.3.3 Algorithmic Strategy: Abstraction